

ASSIGNMENT NO # 02

Sub: Internet of Things (Code: 19EC70E02T)
Program: B.Tech, 7th Semester (Batch: 2022)

Due Date: 3-11-2025
Full Mark: 50

Q. No	CO	Level	Answer all Questions. Each Question Carries 5 Mark BTL: (1) Remembering (2) Understanding (3) Applying (4) Analyzing (5) Evaluating (6) Creating	5 X 10
1	3	2	i. Write a python program Python Program for controlling LED on / off from Raspberry Pi ii. Write the program to interfacing a LDR with Raspberry Pi	
2	3	2	i. How is Raspberry Pi different from desktop computer? ii. What is use od GPIO Pins? iii. Explain the concepts involved in Raspberry Pi	
3	3	3	Illustrate various steps involved in IoT system design methodology. Describe process specification step for home automaton IoT system	
4	3	2	Write a case study on Explain with example IoT based smart home automation	
5	3	2	What is Industry 4.0, and how does it differ from previous industrial revolutions? Explain Or Discuss the principles of Industry 4.0, including Interconnection, Information Transparency, Technical Assistance, and Decentralized Decisions.	
6	4	2	Compare and contrast IoT and IoE (Internet of Everything), focusing on the differences in their scope, applications, and the key technologies that drive them.	
7	4	2	i. What is role of Big data in IoT? Name different types of Big Data. ii. Differentiate between structured and unstructured data in the context of IoT applications	
8	4	2	Explain 7 V' Characteristics of Big Data	
9	4	3	Discuss the role of Big Data analytics in IoT and explain how each type of analysis can be applied in an IoT system.	
10	4	2	Explain the working of RFID technology in detail, including its use in IoT systems and the differences between active, passive, and semi-passive RFID tags.	

*CO3:Students are able to understand the applications area of IoT, IoT enabling technologies, characteristics of IoT systems and IoT levels and Domain-specific IoTs.

*CO4:Students are able to realize the revolutions of Internet in mobile devices, cloud and different sensor networks and apply them in IoT applications.